V9240 Application Note

Frequency and Phase Measurement

Version: **0.1**
Release Date: **June 30, 2017**

Specifications are subject to be changed without notice.

Revision History

Date	Version	Description
2017.06.30	0.1	Initial release

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1. Frequency Measurement

The system samples the voltage signal at 6400Hz and makes a positive zero crossing judgement for the sampling point. Each cycle (20ms) outputs a frequency measurement (ie, the number of samples between the two positive zeroes), stored in the instantaneous frequency value register (FREQINST, 0x00CB). In order to improve the accuracy of the frequency measurement, the instantaneous frequency value is accumulated 16 times to obtain the average frequency value register (FRQAVG, 0x00D2).

$$f' = \frac{6400 \times 16}{FRQAVG}$$ Equation 1-1

f'：Signal frequency value（Frequency value before calibration）;

FRQAVG: Value of average frequency value register (0x00D2);

RC clock offset rate:

$$\mathrm{E} = \frac{T8BAUD}{\frac{8 * 3276800}{K'}} = \frac{K' * T8BAUD}{26214400}$$ Equation 1-2

E：RC clock offset rate;

T8BAUD: Value of register (0x00E0);

K：Standard baud rate（For example: 2400 or 4800）;

K'：Actual baud rate;

RC clock offset calculation frequency can be calculated according to the following equation:

$$f = f' \times E = 0.00390625 \times K' \times \frac{T8BAUD}{FRQAVG}$$ Equation 1-3

f：Actual frequency value;

K': Actual baud rate;

T8BAUD: Value of register (0x00E0);

FRQAVG: Value of average frequency value register (0x00D2);

2. Reverse Phase Sequence Judgement

Write a non-zero value to the register (0x0198) to initiate a phase measurement. Start counting when the UART communication parsing a phase measurement command. Not stop counting until a positive zero crossing event is detected. This counting value should be written into the phase register PHDAT (0x00DE). (Standard one cycle is 20ms with counting 65536 times.)

For example, please follow the steps to do the reverse phase sequence judgment for the three-phase meter:

1. MCU sending the broadcast write instructions to V9240 indicates that writing any non-zero value to the register (0x0198). A, B, and C phases with V9240s should proceed the counting at the same time.
2. Wait for at least one cycle (20ms). Read the phase register PHDAT (0x00DE) from A, B, and C phase with V9240s respectively.
3. According to the data in the phase register PHDAT (0x00DE), judge whether the phase is in the reverse phase sequence or not. There are three kinds of possibility:

1) The three sets of phase data（A：11604，B：34148，C：55327；A<B<C） can be read to judge the phase sequence. If the phase sequence is ABC, it is a positive phase sequence. The three phase data（A：11604，B：55327，C：34148； A<C<B）can be read to judge the phase sequence. If the phase sequence is ACB, it is a reverse phase sequence.

2) The three sets of phase data（A：66708，B：21023，C：43259；B<C<A） can be read to judge the phase sequence. If the phase sequence is BCA, it is a positive phase sequence. The three phase data（A：66708，B：43259，C：21023；B<A<C）can be read to judge the phase sequence. If the phase sequence is BAC, it is a reverse phase sequence.

3) The three sets of phase data（A：22896，B：45269，C：487；C<A<B）can be read to judge the phase sequence. If the phase sequence is CAB, it is a positive phase sequence. The three phase data（A：45269，B：22896，C：487；C<B<A）can be read to judge the phase sequence. If the phase sequence is CBA, it is a reverse phase sequence.

3. Phase Measurement

1. To get RC offset rate

There is a built-in high-frequency RC oscillator in the metering chip. By default, it operates in a system with the fundamental frequency of 50Hz to generate a 3.2MHz RC clock. The UART communication of the metering chip adopts the baud rate adaptive design method. In order to ensure the synchronization, the internal clock counting corresponding to the bit width of 8 bits and one bit for the last UART communication is provided in the metering chip. It is used for calculating the amount of RC variation.

Example: A phase T8BAUD register (0x00DF) is 5588. B phase T8BAUD register (0x00DF) is 5542. C phase T8BAUD register (0x00DF) is 5458. The ideal baud rate issued by MCU is 4877. According to the Equation 1- 2, we can get the RC offset rate. E_A =1.039607086。 E_B =1.031049118， E_C =1.015421524

2. To get bandpass filter coefficient calculated by RC offset rate

When the bandpass filter coefficient is 0x806764B6 by default, and the external signal source is 50HZ, the RC offset rate will be obtained to calculate the updated band-pass filter coefficient via Equation 1-4.

$$BPFPARA' = -2143289344 * COS(\frac{6.2831852 * T}{6400 * E}) \qquad \text{Equation 1-4}$$

$BPFPARA'$: Updated bandpass filter coefficient

T: Standard frequency value

E：RC clock offset rate

For example: Using the RC offset rate: E_A =1.039607086. E_B =1.031049118. E_C =1.015421524. When T=50HZ, the RC offset rates can be used in Equation 1-4 to obtain the following:

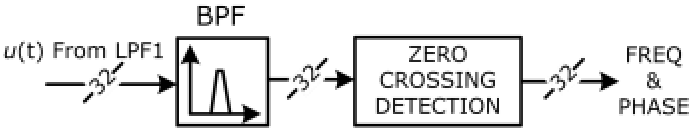
$BPFPARA'_A$ =-2140900590=0X80647312

$BPFPARA'_B$ =-2140860778=0X80650E96

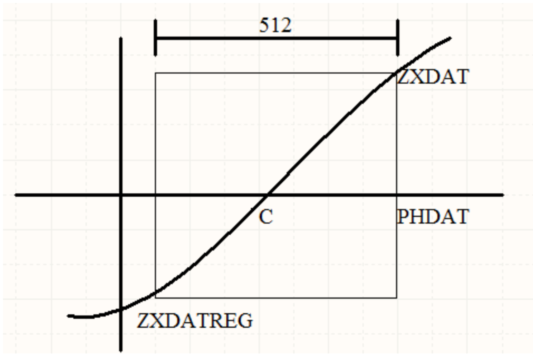
$BPFPARA'_C$ =-2140785465=0X806634C7

And write into the bandpass filter coefficient (0x0107) registers of metering chips.

3. Calculate each phase interpolation



The metering chip supports the voltage phase measurement function. The signal processing is shown in the figure. The working principle is that when the UART communication is analyzed as a phase measurement command, the local high frequency RC clock (typically 3.2MHz) counting will be used until the positive zero-crossing event occurs. The counting will stop, and the counting value will be written into the phase register PHDAT (0x00DE). The two voltage samples ZXDATREG (0x00DC) and ZXDAT (0x00DD) before and after the positive zero-crossing will be recorded, so that the users can do the interpolation operation to get a higher precision phase value. The calculation equation is shown in Figure 1-5.



$$C = PHDAT - \left(\frac{512 * ZXDAT}{ZXDAT - ZXDATREG}\right)$$ Equation 1-5

C : Phase interpolation

PHDAT：Phase register（0x00DE）

ZXDAT: Current voltage zero-crossing sampling value（0x00DD）

ZXDATREG: Previous voltage zero-crossing sampling value（0x00DC）

For example: Read the value of each PHDAT, ZXDAT, and ZXDATREG

$PHDAT_A$ =66708; $ZXDAT_A$ =666084; $ZXDATREG_A$ =-1219397;

$PHDAT_B$ =21023; $ZXDAT_B$ =112947; $ZXDATREG_C$ =-1793406;

$PHDAT_C$ =43259; $ZXDAT_C$ =1502121; $ZXDATREG_C$ =-423189;

Get into the Equation 1-5 to receive C_A =66527. 12573 , C_B =20992. 66519， C_C =42859. 53916

4. Calculate the theoretical value via each phase interpolation

$$C' = \frac{C}{E}$$ Equation 1-6

C'： Phase interpolation theoretical value

C： Phase interpolation calculated by Equation 1-5

E：RC clock offset rate

For example： Get C_A =66527.12573 ，C_B =20992.66519， C_C =42859.53916， E_A =1.039607086。E_B =1.031049118， E_C =1.015421524 into Equation 1-6 to receive

C'_A =63992.56662; C'_B =20360.48993; C'_C =42208.61794;

5. Phase angle calculated by the theoretical phase interpolation

In the case of 50HZ, a cycle is of 65536 points and corresponds to the phase angle of 360 degrees. The number of points for a cycle is related to the actual frequency.

$$D = \frac{6400*512}{\mathrm{f}}$$ Equation 1-7

D: Number of points for a cycle

f：Actual frequency, received according to Equation 1-3

The phase angle equation will be as below:

$$A = \frac{C'_{(B,C)} - C'_A}{D} * 360$$ Equation 1-8

When calculating the phase angle, if the value of the A-phase register PHDAT (0x00DE) is greater than B's or C's, it is necessary to subtract one cycle of the A-phase interpolation for the calculation.

$$A = \frac{C'_{(B,C)} - (C'_A - D)}{D} * 360$$ Equation 1-9

For example: When f=50hz, get into the Equation 1-7 to receive the points of a cycle: D=65536,

C'_A =63992.56662; C'_B =20360.48993; C'_C =42208.61794;

Then, via B phase angle with Equation 1-9 to receive the angle A=120.3218443

Via C phase angle with Equation 1-9 to receive angle A=240.337196

4. Attached Data Records

正向序	50HZ	1	2	3	4	5	6	7	8	9	10
	A	0	0	0	0	0	0	0	0	0	0
	B	119.3	119.1	118.7	119	119	119.2	119.2	119.2	119.1	119.4
	C	240.2	240.2	240.3	240.2	240.4	240.3	240.4	240.3	240.5	240.6
	52.5HZ	1	2	3	4	5	6	7	8	9	10
	A	0	0	0	0	0	0	0	0	0	0
	B	119.3	119.4	119.4	119.3	119.3	119.4	119.5	119.3	119.3	119.3
	C	239.8	239.5	239.6	239.7	239.6	239.9	239.9	239.6	239.6	239.4
	47.5HZ	1	2	3	4	5	6	7	8	9	10
	A	0	0	0	0	0	0	0	0	0	0
	B	119.7	119.8	119.7	119.8	119.6	119.8	119.5	119.6	119.7	119.7
	C	240.6	240.7	240.7	240.6	240.7	240.4	240.7	240.5	241.2	240.5
逆相序（B跟C对换）	50HZ	1	2	3	4	5	6	7	8	9	10
	A	0	0	0	0	0	0	0	0	0	0
	B	239.8	239.4	239.2	239.3	239.6	239.3	239.2	239.2	239.5	239.5
	C	120.3	120	119.9	119.9	119.8	119.8	119.9	119.9	120.1	120
	52.5HZ	1	2	3	4	5	6	7	8	9	10
	A	0	0	0	0	0	0	0	0	0	0
	B	239.2	239.4	239.5	239.5	239.5	239.3	239.4	239.4	239.4	239.3
	C	119.3	119.4	119.4	119.4	119.6	119.4	119.4	119.5	119.4	119.4
	47.5HZ	1	2	3	4	5	6	7	8	9	10
	A	0	0	0	0	0	0	0	0	0	0
	B	240.2	240.2	240.2	240.3	240.3	240.4	240.3	240.2	240.2	240.2
	C	120.2	120.5	120.7	120.6	120.6	120.9	120.7	120.3	120.6	120.7

正相序	60HZ	1	2	3	4	5	6	7	8	9	10
	A	0	0	0	0	0	0	0	0	0	0
	B	120	119.8	120.1	120.1	120.3	119.7	120.1	119.8	119.9	119.9
	C	240	239.5	240	239.8	240.4	239.7	239.8	239.5	239.6	239.8
	62.5HZ	1	2	3	4	5	6	7	8	9	10
	A	0	0	0	0	0	0	0	0	0	0
	B	119.8	120.1	120.4	119.8	119.8	120.2	119.7	119.8	119.7	119.6
	C	239.6	239.8	240.5	239.9	239.9	240.3	239.8	239.6	239.6	240.5
	57.5HZ	1	2	3	4	5	6	7	8	9	10
	A	0	0	0	0	0	0	0	0	0	0
	B	119.6	119.6	119.7	119.8	119.6	119.6	119.9	119.6	119.6	119.7
	C	239.8	239.6	240.1	239.4	239.5	239.6	240.1	239.7	240	239.6
逆相序（B跟C对换）	60HZ	1	2	3	4	5	6	7	8	9	10
	A	0	0	0	0	0	0	0	0	0	0
	B	240.1	240.1	239.8	240	239.4	240.1	240.1	240.2	240.1	240.1
	C	119.7	199.9	119.6	119.5	119.4	119.6	119.5	119.5	119.5	119.8
	62.5HZ	1	2	3	4	5	6	7	8	9	10
	A	0	0	0	0	0	0	0	0	0	0
	B	239.7	240.2	240.1	239.8	239.7	240.2	240.1	240.2	239.7	239.9
	C	119.7	120.3	119.6	119.6	119.6	120	119.8	119.6	119.6	119.7
	57.5HZ	1	2	3	4	5	6	7	8	9	10
	A	0	0	0	0	0	0	0	0	0	0
	B	239.9	240.1	240	240	239.9	239.9	240	239.9	239.6	240
	C	119.5	119.7	119.5	119.7	119.6	119.6	119.5	119.6	119.5	119.5